Land 10: Cardia Citadel

Blood and the heart · OpenStax Ch. 18 and 19

Key ideas

Blood composition and functions

- Blood transports gases, nutrients, wastes and hormones; defends against pathogens; and regulates pH, temperature and fluid balance.
- Plasma is about 55 percent of whole blood and is about 92 percent water; formed elements make up the rest.
- Plasma proteins: albumin (most abundant, colloid osmotic pressure), globulins (transport and antibodies), fibrinogen (clotting). Most come from the liver.
- Hematocrit is the percent of blood volume that is red cells, roughly 37 to 47 in females and 42 to 52 in males.
- The buffy coat (under 1 percent) holds leukocytes and platelets.

Hematopoiesis and erythrocytes

- All formed elements arise from hemopoietic stem cells in red bone marrow.
- Erythropoietin from the kidneys (released when oxygen is low) drives red cell production; thrombopoietin from the liver drives platelet production.
- Erythrocytes are biconcave, lack nuclei and mitochondria, and live about 120 days.
- Each hemoglobin has four globin chains and four iron containing hemes, carrying up to four O2.
- Recycling: globin to amino acids, iron to transferrin (stored as ferritin), heme to biliverdin then bilirubin excreted in bile.

Anemias and red cell disorders

- Iron deficiency anemia: small, pale red cells.
- Pernicious anemia: no intrinsic factor, so B12 cannot be absorbed.
- Kidney disease reduces EPO and causes anemia.
- Hemolytic anemia destroys red cells faster than they are replaced and can cause jaundice; sickle cell disease is an inherited hemoglobin defect.
- Polycythemia is an abnormally high red cell count that thickens blood.

Leukocytes and platelets

- Normal leukocyte count is about 5,000 to 10,000 per microliter; higher is leukocytosis, lower is leukopenia.
- Order of abundance: neutrophils, lymphocytes, monocytes, eosinophils, basophils (Never Let Monkeys Eat Bananas).
- Neutrophils are first responders to bacteria; eosinophils fight parasites and moderate allergy; basophils release histamine and heparin.
- Lymphocytes (B, T, NK cells) drive specific immunity; monocytes become macrophages in tissues.
- Platelets are fragments of megakaryocytes and live about 10 days.

Hemostasis and clotting disorders

- Three steps: vascular spasm, platelet plug (von Willebrand factor helps platelets stick to collagen), coagulation.
- Extrinsic pathway starts with tissue factor; intrinsic pathway starts with factor XII; both converge on factor X in the common pathway.
- Prothrombinase converts prothrombin to thrombin; thrombin converts fibrinogen to fibrin. Calcium and vitamin K are essential.
- Fibrinolysis: plasminogen becomes plasmin, which dissolves fibrin.
- Hemophilia is an X linked clotting factor deficiency; thrombocytopenia is a low platelet count; thrombosis is an unwanted clot and an embolus is one that travels.

Blood typing and transfusion

- Type A cells carry A antigens and plasma has anti B; type B is the reverse; AB has both antigens and no ABO antibodies; O has neither antigen and both antibodies.
- O negative is the universal donor of red cells; AB positive is the universal recipient.
- Rh antibodies form only after exposure to Rh D antigen.
- Hemolytic disease of the newborn: an Rh negative mother sensitized by an earlier Rh positive fetus makes anti D IgG that crosses the placenta.
- RhoGAM given in pregnancy and after delivery prevents sensitization.

Heart anatomy and circulation

- The heart lies in the mediastinum; the pericardium has a fibrous layer and a serous layer (parietal and visceral) with the pericardial cavity between them.
- Wall layers: epicardium, myocardium, endocardium.
- Right side pumps to the pulmonary circuit; left side pumps to the systemic circuit with a thicker wall to generate higher pressure.
- AV valves (tricuspid on the right, mitral on the left) are anchored by chordae tendineae; semilunar valves guard the pulmonary trunk and aorta.
- Left coronary artery gives the anterior interventricular and circumflex arteries; the right coronary artery supplies much of the right heart; veins drain to the coronary sinus and then the right atrium.

Conduction, action potentials and ECG

- Pathway: SA node, internodal paths, AV node (delay), AV bundle, bundle branches, Purkinje fibers.
- SA node cells have a prepotential from slow Na+ leak; without nervous input they fire about 100 times per minute.
- Contractile cells: rapid Na+ depolarization, then a plateau from slow Ca2+ influx, giving a long refractory period that prevents tetany.
- Intercalated discs contain gap junctions (electrical coupling) and desmosomes (mechanical strength).
- ECG: P wave is atrial depolarization, QRS is ventricular depolarization (hides atrial repolarization), T wave is ventricular repolarization; PR interval about 0.12 to 0.20 seconds.

Cardiac cycle, output and regulation

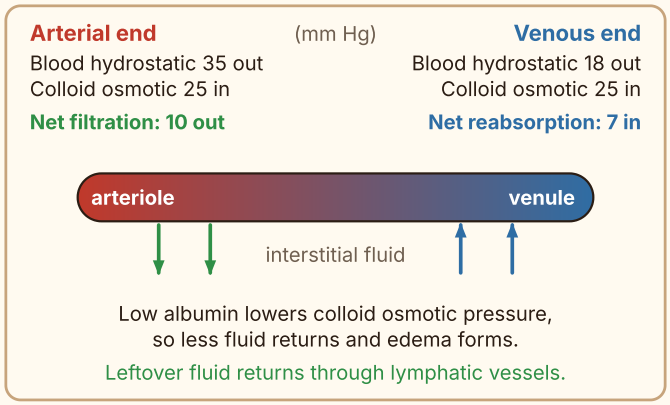
- S1 is AV valve closure at the start of ventricular systole; S2 is semilunar valve closure at the start of diastole. Isovolumic phases occur with all valves closed.
- Cardiac output equals heart rate times stroke volume, about 5.25 L/min at rest (75 bpm times 70 mL).
- Stroke volume equals end diastolic volume (about 130 mL) minus end systolic volume (about 60 mL).
- Preload (Frank Starling stretch) and contractility raise stroke volume; afterload lowers it.
- Sympathetic input, epinephrine and thyroid hormone raise heart rate; vagal acetylcholine lowers it; high potassium risks arrhythmia and arrest.
- Left sided failure causes pulmonary congestion; right sided failure causes systemic edema; a myocardial infarction is tissue death from blocked coronary flow.

Villager wisdom

- Down in the Citadel cellars we say the kidneys watch the oxygen. When it runs low they send erythropoietin to the marrow, and new red cells march out.
- Stopping a bleed takes three steps, traveler: the vessel squeezes, platelets pile into a plug, then fibrin threads stitch it tight.
- Know a blood type by its antigens. Type A cells wear A antigens and the plasma carries anti B antibodies, so type B blood is a foe.
- The SA node is the drummer of the heart. The AV node holds the beat a moment so the atria can finish filling the ventricles.
- Cardiac output is heart rate times stroke volume. Stroke volume rises with more preload or contractility and falls when afterload climbs.

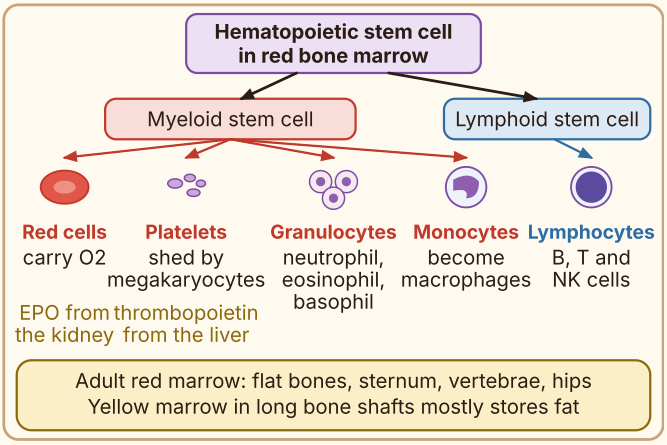
Diagrams

Capillary filtration and reabsorption



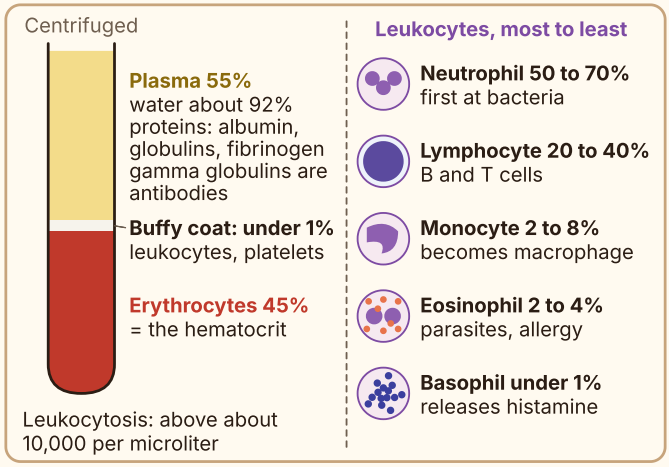
At the arterial end blood hydrostatic pressure (35 mm Hg) beats colloid osmotic pressure (25) so fluid filters out; at the venous end (18 versus 25) fluid returns.

Hematopoiesis



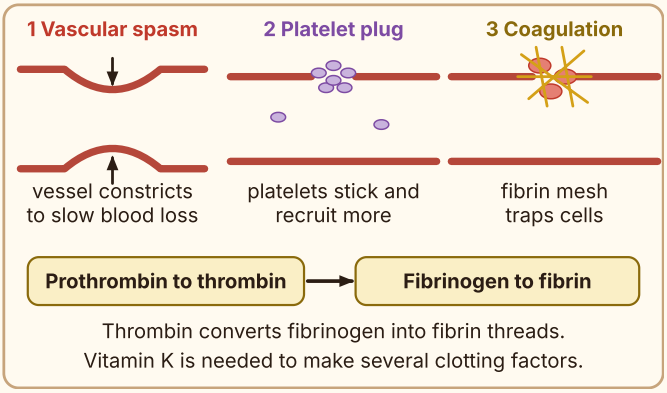
All formed elements come from hematopoietic stem cells in red bone marrow; erythropoietin from the kidney drives red cell production and thrombopoietin from the liver drives platelet production.

Composition of blood



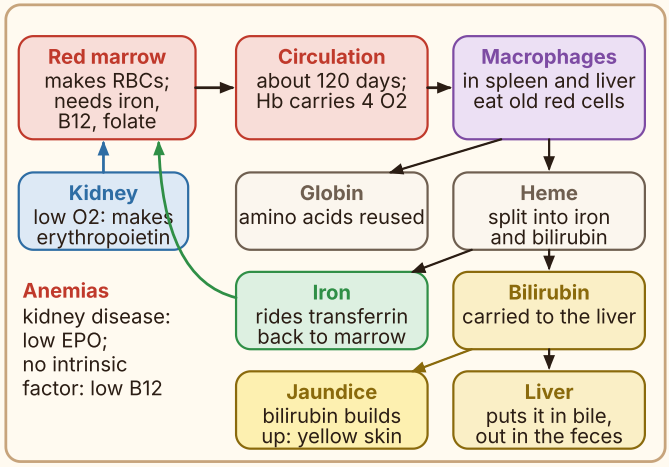
Plasma is about 55 percent of blood and mostly water; erythrocytes make up about 45 percent (the hematocrit); the thin buffy coat holds leukocytes and platelets, and neutrophils are the most numerous leukocyte.

Hemostasis



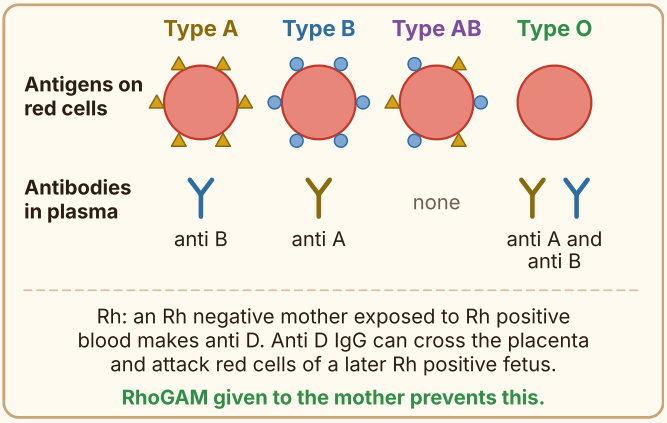
Hemostasis runs vascular spasm, platelet plug, then coagulation, where thrombin turns fibrinogen into a fibrin mesh; vitamin K is needed to make several clotting factors.

Life cycle of a red blood cell



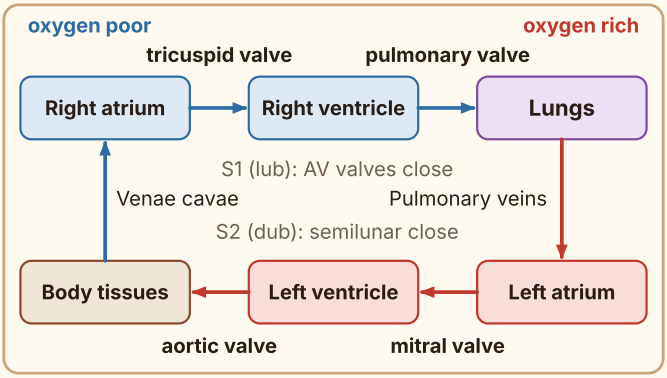
Red cells circulate about 120 days; macrophages recycle them, iron returns to the marrow on transferrin, and bilirubin goes to the liver and out in bile, so a buildup of bilirubin causes jaundice.

ABO and Rh blood groups



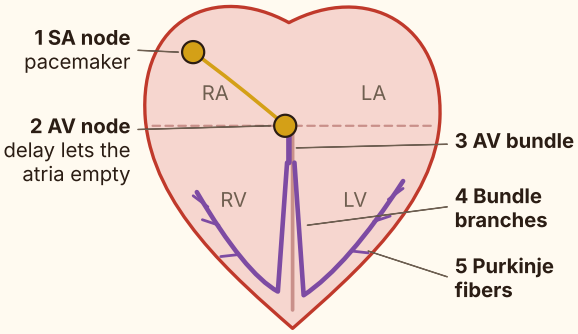
Plasma holds antibodies against the ABO antigens a person lacks, so type A blood carries anti B and attacks type B donor cells.

Path of blood through the heart



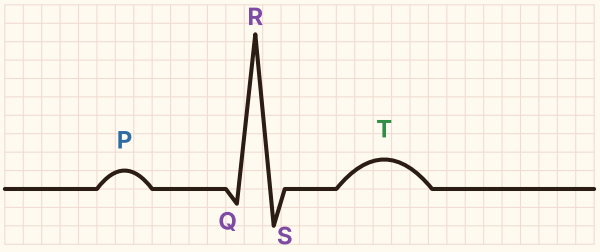
Blood passes right atrium, tricuspid valve, right ventricle, pulmonary valve, lungs, left atrium, mitral valve, left ventricle, aortic valve, body; AV valve closure makes S1.

Cardiac conduction system



The SA node sets the pace; the AV node delays the signal so the atria finish emptying, then the AV bundle, bundle branches and Purkinje fibers spread it through the ventricles.

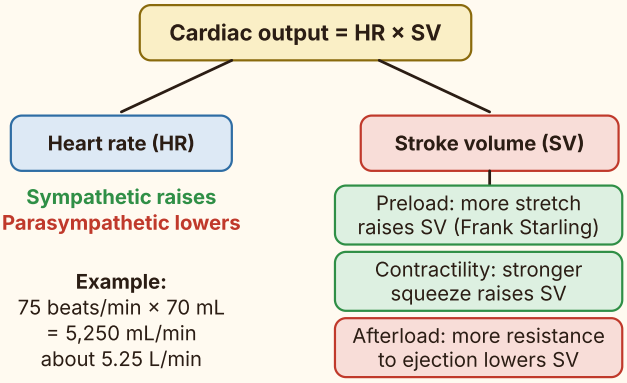
The ECG



P wave: atria depolarize
QRS complex: ventricles depolarize
T wave: ventricles repolarize
Atrial repolarization is hidden in the QRS.

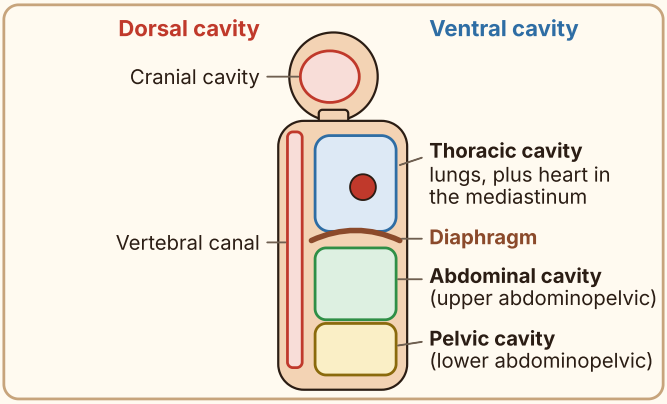
The P wave is atrial depolarization, the QRS complex is ventricular depolarization (hiding atrial repolarization), and the T wave is ventricular repolarization.

Cardiac output



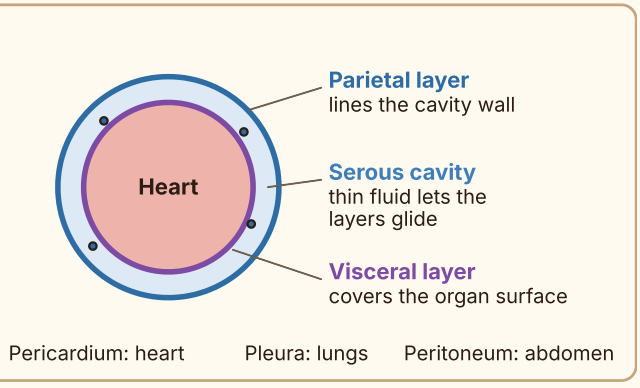
Cardiac output is heart rate times stroke volume; more preload or contractility raises stroke volume, and higher afterload lowers it.

Body cavities



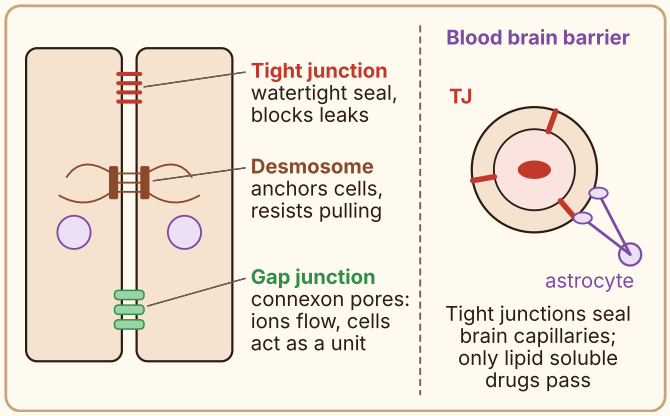
The dorsal cavity holds the brain and spinal cord; the ventral cavity is split by the diaphragm into the thoracic cavity above and the abdominopelvic cavity below.

Serous membranes



A serous membrane has a visceral layer on the organ and a parietal layer on the cavity wall; serous fluid between them lets the layers glide without friction.

Cell junctions and the blood brain barrier



Tight junctions seal neighboring cells, desmosomes rivet them against pulling, and gap junctions (connexons) let ions pass so cells act together; tight junctions in brain capillaries form the blood brain barrier.

Self check

1. **The first heart sound (S1, the lub) is caused mainly by:**
    A. Closure of the semilunar valves
    B. Opening of the AV valves
    C. Closure of the AV valves
    D. Blood rushing into the atria
2. **The heart is located within which region of the thoracic cavity?**
    A. Pleural cavity
    B. Peritoneal cavity
    C. Abdominal cavity
    D. Mediastinum
3. **A patient has a dangerously high blood potassium level. Which cardiac effect is the greatest concern?**
    A. Stronger contractions from extra calcium
    B. A steady rise in stroke volume
    C. Arrhythmias and possible cardiac arrest
    D. Faster conduction with no risk
4. **During an extremely rapid tachycardia, cardiac output can fall even though heart rate is high. Why?**
    A. Systole lasts longer than normal
    B. Afterload drops to zero
    C. Contractility always decreases with rate
    D. Shortened diastole reduces ventricular filling
5. **During fibrinolysis, which enzyme digests the fibrin of a clot?**
    A. Thrombin
    B. Prothrombinase
    C. Fibrinogen
    D. Plasmin
6. **On a normal ECG, the QRS complex represents:**
    A. Atrial depolarization
    B. Ventricular depolarization
    C. Ventricular repolarization
    D. Firing of the SA node
7. **The plateau of the contractile cardiac muscle action potential is caused mainly by:**
    A. Rapid Na+ influx through fast channels
    B. K+ influx through open channels
    C. Slow Ca2+ influx balancing K+ outflow
    D. Chloride entry hyperpolarizing the cell

8. **The biconcave disc shape of an erythrocyte mainly benefits the cell by:**
    A. Increasing surface area for gas diffusion
    B. Making room for a large nucleus
    C. Storing iron for later use
    D. Allowing the cell to divide quickly
9. **Which layer of the heart wall is made mostly of cardiac muscle?**
    A. Epicardium
    B. Endocardium
    C. Myocardium
    D. Fibrous pericardium
10. **Serous fluid that lets the heart beat with little friction is found in the:**
    A. Space between fibrous and parietal layers
    B. Myocardium
    C. Pericardial cavity between serous layers
    D. Lumen of the coronary sinus
11. **A patient with chronic kidney disease has a low hematocrit despite normal iron, B12 and folate. The most likely cause is:**
    A. Reduced erythropoietin production
    B. Increased thrombopoietin release
    C. Destruction of red cells by antibodies
    D. Loss of intrinsic factor
12. **Most plasma proteins are made by the liver. Which plasma proteins are produced by plasma cells instead?**
    A. Albumin
    B. Fibrinogen
    C. Alpha and beta globulins
    D. Gamma globulins (antibodies)

Answer key

1. **C.** Closure of the AV valves. S1 occurs as the ventricles begin to contract and rising pressure snaps the tricuspid and mitral valves shut. Closure of the semilunar valves makes S2, the dub, at the end of ventricular systole.
2. **D.** Mediastinum. The heart sits in the mediastinum, between the lungs and behind the sternum, with its apex tilted toward the left. The pleural cavities surround the lungs, not the heart.
3. **C.** Arrhythmias and possible cardiac arrest. Hyperkalemia alters the resting membrane potential of cardiac cells, disrupting conduction and causing dangerous arrhythmias that can end in cardiac arrest. Stronger contractions are linked to high calcium, not high potassium.
4. **D.** Shortened diastole reduces ventricular filling. Fast rates mostly shorten diastole, so the ventricles fill less, end diastolic volume and stroke volume drop, and the product of rate and volume can fall. Systole shortens only slightly, it does not lengthen.
5. **D.** Plasmin. Plasminogen is converted to plasmin, which breaks fibrin into fragments and dissolves the clot as the vessel heals. Thrombin does the opposite, converting fibrinogen into fibrin.

6. **B.** Ventricular depolarization. The QRS complex records the rapid depolarization of the large ventricular muscle mass, which triggers ventricular contraction. Ventricular repolarization is the T wave, while the P wave shows atrial depolarization.
7. **C.** Slow Ca2+ influx balancing K+ outflow. During the plateau, slow calcium channels let Ca2+ enter while some K+ leaves, holding the membrane depolarized and lengthening the refractory period. Rapid Na+ influx causes the initial depolarization, not the plateau.
8. **A.** Increasing surface area for gas diffusion. The thin, dimpled shape gives a large surface area relative to volume and lets the cell bend through narrow capillaries. Mature erythrocytes have no nucleus and cannot divide.
9. **C.** Myocardium. The myocardium is the thick muscular middle layer that does the pumping. The epicardium is the outer serous layer (visceral pericardium), and the endocardium lines the chambers.
10. **C.** Pericardial cavity between serous layers. The pericardial cavity lies between the parietal and visceral serous layers and holds a small amount of lubricating fluid. The fibrous pericardium is fused to the parietal layer, so there is no fluid space between them.
11. **A.** Reduced erythropoietin production. Damaged kidneys make less erythropoietin, so the marrow receives less signal to produce red cells. Loss of intrinsic factor would cause B12 deficiency, which is ruled out here.
12. **D.** Gamma globulins (antibodies). Antibodies (immunoglobulins) are made by plasma cells derived from B lymphocytes. Albumin, fibrinogen and the alpha and beta globulins all come from the liver.